

Solution Definition Document

Swiggy Order Analytics / Power BI Project

1. Solution Overview

The solution is an interactive Power BI dashboard built on a simple relational data model consisting of Orders, Customers, and Restaurants. Calculated columns and DAX measures are used to standardize business calculations and provide reusable metrics across visuals.

2. Data Model

Recommended relationships:

- Customers[CustomerID] (1) → Orders[CustomerID] (*)
- Restaurants[RestaurantID] (1) → Orders[RestaurantID] (*)

3. Core Calculated Columns

Net Revenue: $\text{Orders}[\text{OrderValue}] - \text{Orders}[\text{DiscountAmount}] + \text{Orders}[\text{DeliveryFee}]$

Order Month: $\text{FORMAT}(\text{Orders}[\text{OrderDate}], "MMM YYYY")$

Delivery Speed: $\text{IF}(\text{Orders}[\text{DeliveryTimeMin}] \leq 30, "Fast", "Slow")$

Discount %: $\text{DIVIDE}(\text{Orders}[\text{DiscountAmount}], \text{Orders}[\text{OrderValue}], 0)$

Day Type: $\text{IF}(\text{WEEKDAY}(\text{Orders}[\text{OrderDate}], 2) > 5, "Weekend", "Weekday")$

4. Core DAX Measures

Total Revenue: $\text{Total Revenue} = \text{SUM}(\text{Orders}[\text{Net Revenue}])$

Total Orders: $\text{Total Orders} = \text{COUNTROWS}(\text{Orders})$

Avg Order Value: $\text{Avg Order Value} = \text{DIVIDE}([\text{Total Revenue}], [\text{Total Orders}], 0)$

Cancellation Rate: $\text{Cancellation Rate} = \text{DIVIDE}(\text{CALCULATE}([\text{Total Orders}], \text{Orders}[\text{OrderStatus}] = "Cancelled"), [\text{Total Orders}], 0)$

Avg Delivery Time: $\text{Avg Delivery Time} = \text{AVERAGE}(\text{Orders}[\text{DeliveryTimeMin}])$

5. Dashboard Visuals

Visual	Fields	Purpose
KPI Cards	Total Revenue; Total Orders; Avg Order Value; Cancellation Rate	High-level performance

Bar Chart	RestaurantName + Total Revenue	Top restaurants by net revenue
Line Chart	Order Month + Total Revenue	Revenue trend over time
Donut Chart	PaymentMethod + Total Orders	Order share by payment method
City Performance	City + Total Revenue + Total Orders	Compare geographic performance
Stacked Bar	Cuisine + OrderStatus + Total Orders	Order status split by cuisine
Scatter	Rating + Avg Order Value + Total Orders as size	Explore rating/order-value relationship

6. Interactive Filters

- PaymentMethod
- City
- OrderStatus
- Cuisine
- OrderDate

7. Business Value

- Provides a single view of order and revenue performance.
- Helps identify high-contributing restaurants and cities.
- Highlights cancellation and delivery-performance patterns.
- Shows payment-method usage and cuisine-level order-status distribution.
- Supports interactive exploration through slicers and cross-filtering.

8. Implementation Notes

- Create Net Revenue as a calculated column, not as a measure, because it is calculated row by row.
- Create aggregate metrics such as Total Revenue and Total Orders as measures.
- Format Cancellation Rate and Discount % as percentages.
- Use the Orders table for transaction measures and Customers/Restaurants for descriptive dimensions.
- For reliable time-intelligence calculations, a dedicated Calendar table can be added later.